

Analytical Problem Solving1) Gray level Transformation:

a) Explain the purpose of grey level transformation in image formation

Grey level transformation is a point processing technique in which every pixel intensity of an image is modified according to a mathematical function

Purpose

- Improves the brightness of an image
- Enhances the contrast b/w objects and background.
- Highlights important object details

$$S = 255 - r$$

$r$  = original pixel intensity

$S$  = Transformed pixel intensity

b) Apply the negative transformation

Given,

$$r = 50, 100, 150, 200$$

$$S = 255 - r$$

$$\text{For } r = 50$$

$$S = 255 - 50 \\ = 205$$

$$\text{For } r = 100$$

$$S = 255 - 100 \\ S = 155$$

$$\text{For } r = 150$$

$$S = 255 - 150$$

$$= 105$$

$$\text{For } r = 200$$

$$S = 255 - 200$$

$$= 55$$

- Dark pixels becomes bright
- Bright pixels become dark

## 2) Histogram Equalization

a) Explain why histogram equalization is used

Histogram equalization is a image enhancement technique used to improve image contrast by redistributing the pixel intensity values over the entire intensity range.

Given Histogram,

| Intensity | Frequency |
|-----------|-----------|
| 0         | 4         |
| 1         | 6         |
| 2         | 10        |
| 3         | 6         |

Cumulative Freq

$$\begin{aligned}
 N_1 &= 4 \\
 N_2 &= 4 + 6 = 10 \\
 N_3 &= 10 + 6 + 4 = 20 \\
 N_4 &= 10 + 6 + 6 + 4 = 26
 \end{aligned}$$

$N = 26$

b) Compute CDF

$$CDF = \frac{\text{Cumulative Freq}}{26}$$

| Intensity | CDF   |
|-----------|-------|
| 0         | 0.154 |
| 1         | 0.385 |
| 2         | 0.769 |
| 3         | 1     |

PDF-9

c) Determine Equalized Intensity

Given  $L = 4$

Transformed Formula =  $S = (L-1) \times CDF$

$i=0 \Rightarrow 3 \times 0.154 = 0.462 \approx 0$

$i=1 \Rightarrow 3 \times 0.385 = 1.155 \approx 1$

$$i=2 \Rightarrow 3 \times 0.769 = 2.307 \approx 2$$

$$i=3 \Rightarrow 3 \times 1 = 3$$

### 3) Interpretation

- Pixel values are and the contrast is improved by spreading intensity values more uniformly.

### 3) Gaussian smoothing

#### a) Purpose

Gaussian filtering reduces image noise while preserving edges better than averaging filtering.

#### b) Calculation

Kernel:

$$\frac{1}{16} \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

Image:

$$\begin{bmatrix} 10 & 20 & 10 \\ 20 & 40 & 20 \\ 10 & 20 & 10 \end{bmatrix}$$

$$\frac{(10+20+10+20+40+20+10+20+10)}{16} = \frac{360}{16} = 22.5$$

New center pixel = 22.5  $\approx$  23

Interpretation: Noise is reduced and image became smoother

#### 4) Edge detection using sobel operator

##### a) Purpose

Edge detection identifies object boundaries by detecting sudden intensity changes

##### b) Calculation

sobel mask:

$$\begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix}$$

Image:

$$\begin{bmatrix} 10 & 10 & 10 \\ 20 & 20 & 20 \\ 30 & 30 & 30 \end{bmatrix}$$

$(-10 - 20 - 10) + (30 + 60 + 30) = -40 + 120 = 80$

Gradient = 80

Interpretation: A strong horizontal edge is detected

#### 5) Contrast stretching

##### a) Purpose

contrast stretching increase the intensity range to improve image contrast.

## b) calculation

Formula:

$$S = \frac{(r-50)}{150-50} \times 255$$

For  $r=50$

$$S=0$$

For  $r=100$

$$S = \frac{50}{100} \times 255 = 127.5 \approx 128$$

For  $r=150$

$$S = \frac{100}{100} \times 255 = 255$$

| Original | Transformed |
|----------|-------------|
| 50       | 0           |
| 100      | 128         |
| 150      | 255         |

Interpretation:

The intensity range is expanded from 50-150 to 0-255, improving image contrast